SUMMATIVE ASSESSMENT – 1 (SA-1) MODEL PAPER – SET 1

CLASS IX · MATHEMATICS · Semester-1 · AP SCERT 2026-27 · SSC 100-Mark Pattern

BLUEPRINT – SA-1 MODEL PAPER – SET 1

SUBJECT: MATHEMATICS	CLASS: IX	TIME: 3 Hrs 15 Min	MAX. MARKS: 100

Syllabus: Class IX Mathematics, Semester-1 (AP SCERT 2026-27) – 1. Number Systems, 2. Polynomials, 3. Coordinate Geometry, 4. Linear Equations in Two Variables, 5. Introduction to Euclid's Geometry, 6. Lines and Angles. Pattern: SSC 100-mark Summative Assessment blueprint with Academic Standards (Awareness 60%, Sensitivity 20%, Creativity 20%).

1. Weightage to Academic Standards (Ability)

ABILITY	AWARENESS (60%)	SENSITIVITY (20%)	CREATIVITY (20%)	TOTAL
MARKS	60	20	20	100

2. Weightage to Form of Questions

FORM OF QUESTION	Essay / Long Ans.	Short Answer	Very Short Ans.	Objective	TOTAL
No. of questions	5	8	8	12	33
Marks allotted	8	4	2	1	100
Estimated time	70 min	60 min	30 min	20 min	180 + 15 min

3. Ability-wise Distribution of Questions

ABILITY	1 Mark	2 Marks	4 Marks	8 Marks	Total Marks
Awareness	4	4	6	3	60
Sensitivity	4	2	1	1	20
Creativity	4	2	1	1	20
Total Questions	12	8	8	5	100

4. Chapter-wise Weightage

S.No	UNIT / CHAPTER	MARKS
1	Number Systems	1+1+2+2+4+4+8+(8) = 22 + (8)
2	Polynomials	1+1+2+4+4+8+(8) = 20 + (8)
3	Linear Equations in Two Variables	1+1+2+4+8+(8) = 16 + (8)
4	Coordinate Geometry	1+1+2+4+8 = 16
5	Introduction to Euclid's Geometry	1+1+2+4+(8) = 8 + (8)
6	Lines and Angles	1+1+2+2+4+8+(8) = 18 + (8)
	TOTAL	**100 + (40)**

Note: Marks within ( ) are the internal-choice (OR) questions of Section IV. The 100 marks are distributed over the six Semester-1 chapters; the ability-wise (60 : 20 : 20) and form-wise weightage follow the SSC blueprint pattern.

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5. Question-wise Chapter Distribution and Domain (Questions 1 to 33)

Q.No	Chapter / Topic	Domain	Marks
1	Number Systems	Awareness	1
2	Polynomials	Creativity	1
3	Linear Equations in Two Variables	Creativity	1
4	Coordinate Geometry	Creativity	1
5	Introduction to Euclid’s Geometry	Sensitivity	1
6	Lines and Angles	Awareness	1
7	Number Systems	Sensitivity	1
8	Polynomials	Sensitivity	1
9	Introduction to Euclid’s Geometry	Creativity	1
10	Coordinate Geometry	Awareness	1
11	Linear Equations in Two Variables	Awareness	1
12	Lines and Angles	Sensitivity	1
13	Number Systems	Creativity	2
14	Number Systems	Awareness	2
15	Polynomials	Awareness	2
16	Coordinate Geometry	Sensitivity	2
17	Linear Equations in Two Variables	Awareness	2
18	Introduction to Euclid’s Geometry	Sensitivity	2
19	Lines and Angles	Awareness	2
20	Lines and Angles	Creativity	2
21	Number Systems	Awareness	4
22	Number Systems	Awareness	4
23	Polynomials	Awareness	4
24	Polynomials	Awareness	4
25	Linear Equations in Two Variables	Sensitivity	4
26	Coordinate Geometry	Awareness	4
27	Introduction to Euclid’s Geometry	Awareness	4
28	Lines and Angles	Creativity	4
29a / 29b	(a) Number Systems (b) Introduction to Euclid’s Geometry	Creativity	8
30a / 30b	(a) Polynomials (b) Number Systems	Awareness	8
31a / 31b	(a) Linear Equations in Two Variables (b) Polynomials	Sensitivity	8
32a / 32b	(a) Lines and Angles (b) Linear Equations in Two Variables	Awareness	8
33a / 33b	(a) Coordinate Geometry (b) Lines and Angles	Awareness	8

ABILITIES AND THEIR OBJECTIVES AS PER PARAKH NORMS

I) AWARENESS: Knowledge / understanding of the construct / understanding of the process (resembles old objectives Knowledge, Understanding and Application).

II) SENSITIVITY: Conflict resolution / open-mindedness / collaboration (resembles old objective Analysis).

III) CREATIVITY: Generate innovative ideas / flexibility / fluency / exploration / combining ideas and concepts (resembles old objectives Evaluation and Creativity).

NOTE: Overall, the chapter-wise and ability-wise weightage is followed. There is no rigid rule that a question with a particular number must come from a particular chapter.

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SUMMATIVE ASSESSMENT – 1 : MODEL QUESTION PAPER – SET 1

MATHEMATICS – CLASS IX (Semester-1, AP SCERT Syllabus 2026-27)

Time : 3 Hrs 15 Min (15 min reading + 3 hrs writing) **Max. Marks : 100**

General Instructions: *(i) Answer all the questions. (ii) Write the answers neatly in the answer booklet. (iii) The paper has four Sections (I, II, III, IV). (iv) Use of calculators is not permitted. (v) Draw neat, labelled figures wherever necessary.*

SECTION – I 12 × 1 = 12 Marks

Note : *i) Answer all the questions. ii) Each question carries 1 mark. iii) Choose the correct option and write its letter (A/B/C/D) in the answer booklet.*

1. Which of the following is an irrational number? *[Awareness · Number Systems]*

(A) √16 (B) 22/7 (C) √7 (D) 0.25

2. If (x − 2) is a factor of $x^3 - 6x^2 + kx - 8$, then k = *[Creativity · Polynomials]*

(A) 10 (B) 12 (C) 14 (D) 16

3. The graph of the equation 2x + 3y = 6 cuts the y-axis at the point *[Creativity · Linear Equations in Two Variables]*

(A) (0, 3) (B) (0, 2) (C) (2, 0) (D) (3, 0)

4. The point which lies on the y-axis, 5 units below the x-axis, is *[Creativity · Coordinate Geometry]*

(A) (−5, 0) (B) (0, −5) (C) (5, 0) (D) (0, 5)

5. Rahul says 'The whole is greater than the part' is a postulate, but Sita says it is an axiom. Who is correct? *[Sensitivity · Introduction to Euclid's Geometry]*

(A) Rahul (B) Sita (C) Both (D) None

6. The supplement of 70° is *[Awareness · Lines and Angles]*

(A) 20° (B) 110° (C) 290° (D) 100°

7. The value of (√5 + √2)(√5 − √2) is *[Sensitivity · Number Systems]*

(A) 3 (B) 7 (C) √10 (D) 10

8. The zeroes of $p(x) = x^2 - 5x + 6$ are *[Sensitivity · Polynomials]*

(A) 2 and 3 (B) −2 and −3 (C) 1 and 6 (D) −1 and −6

9. Through a given point not on a line, the number of lines parallel to the given line is *[Creativity · Introduction to Euclid's Geometry]*

(A) none (B) exactly one (C) exactly two (D) infinitely many

10. The abscissa of the point (−5, 3) is *[Awareness · Coordinate Geometry]*

(A) 3 (B) −5 (C) 5 (D) −3

11. A linear equation in two variables has *[Awareness · Linear Equations in Two Variables]*

(A) one solution (B) two solutions
(C) no solution (D) infinitely many solutions

12. The angles of a triangle are in the ratio 2 : 3 : 4. The largest angle is *[Sensitivity · Lines and Angles]*

(A) 60° (B) 80° (C) 90° (D) 100°

SECTION – II 8 × 2 = 16 Marks

Note : *i) Answer all the questions. ii) Each question carries 2 marks.*

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13. Find two irrational numbers between 2 and 3. *[Creativity · Number Systems]*

14. Express 0.3636... in the form p/q. *[Awareness · Number Systems]*

15. Find the value of $p(x) = 5x - 4x^2 + 3$ at $x = 0$ and $x = -1$. *[Awareness · Polynomials]*

16. Write the quadrant (or axis) in which each point lies: (−2, 5), (3, −4), (0, −3), (−1, −1). *[Sensitivity · Coordinate Geometry]*

17. Write $3x = 5 - 2y$ in the form $ax + by + c = 0$ and identify a, b, c. *[Awareness · Linear Equations in Two Variables]*

18. Rahim says two distinct lines can have two points in common, but Reshma disagrees. Who is correct? Give reason. *[Sensitivity · Introduction to Euclid's Geometry]*

19. Two supplementary angles are in the ratio 4 : 5. Find the angles. *[Awareness · Lines and Angles]*

20. Two lines AB and CD intersect at O. If $\angle AOC + \angle BOD = 130°$, find $\angle AOD$. *[Creativity · Lines and Angles]*

SECTION – III 8 × 4 = 32 Marks

Note : *i) Answer all the questions. ii) Each question carries 4 marks.*

21. Rationalise the denominators: (i) $1/(\sqrt{7} - \sqrt{6})$ (ii) $1/(\sqrt{5} + \sqrt{2})$. *[Awareness · Number Systems]*

22. Simplify: (i) $2^{2/3} \times 2^{1/3}$ (ii) $16^{3/4}$ (iii) $7^{1/2} \div 7^{1/3}$ (iv) $9^{-3/2}$. *[Awareness · Number Systems]*

23. Factorise: $x^3 - 2x^2 - x + 2$. *[Awareness · Polynomials]*

24. Using suitable identities evaluate: (i) 103 × 107 (ii) $(99)^3$. *[Awareness · Polynomials]*

25. The cost of a notebook is twice the cost of a pen. Write a linear equation in two variables to represent this. Ravi says (5, 10) is a solution and Sita says (4, 6) is a solution, where x = cost of a pen and y = cost of a notebook. Who is correct? *[Sensitivity · Linear Equations in Two Variables]*

26. Plot the points (3, 0), (0, 3), (−3, 0) and (0, −3) on a graph paper, join them in order and name the figure obtained. *[Awareness · Coordinate Geometry]*

27. State Euclid's five postulates. *[Awareness · Introduction to Euclid's Geometry]*

28. The interior angles on the same side of a transversal cutting two parallel lines are in the ratio 2 : 3. Find the angles. *[Creativity · Lines and Angles]*

SECTION – IV 5 × 8 = 40 Marks

Note : *i) Answer all the questions. ii) Each question carries 8 marks. iii) There is an internal choice for each question.*

29.

a) i) Find five rational numbers between 3/5 and 4/5.
ii) Find three irrational numbers between 1/7 and 2/7. *[Creativity · Number Systems]*

OR

b) i) A, B, C are three points on a line and B lies between A and C. Prove that AB + BC = AC using Euclid's axiom.
ii) Prove that every line segment has one and only one mid-point. *[Creativity · Introduction to Euclid's Geometry]*

30.

a) Factorise $2x^3 - 3x^2 - 17x + 30$ using the factor theorem. *[Awareness · Polynomials]*

OR

b) If $x = 3 + 2\sqrt{2}$, find (i) $1/x$ (ii) $x + 1/x$ (iii) $x^2 + 1/x^2$. *[Awareness · Number Systems]*

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31.

a) A shopkeeper says: 'x notebooks at Rs. 15 each and y pens at Rs. 5 each cost Rs. 60 in all.' Write a linear equation in two variables, give four solutions in a table and draw its graph. From the graph, find the number of pens if 2 notebooks are bought. *[Sensitivity · Linear Equations in Two Variables]*

OR

b) Without actual division, prove that $x^3 - 3x^2 - 13x + 15$ is exactly divisible by $x^2 + 2x - 3$. *[Sensitivity · Polynomials]*

32.

a) i) Prove that if two lines intersect, then the vertically opposite angles are equal.
ii) Two lines AB and CD intersect at O. If $\angle AOC = (3x + 10)°$ and $\angle BOD = (5x - 30)°$, find x and all four angles. *[Awareness · Lines and Angles]*

OR

b) Draw the graphs of $x = 3$, $y = -2$ and $x + y = 4$ on the same graph paper. Find the area of the triangle formed by them. *[Awareness · Linear Equations in Two Variables]*

33.

a) Plot the points A(1, 2), B(5, 2), C(5, 5) and D(1, 5) on a graph paper. Join them in order and (i) name the figure, (ii) find the lengths of its sides, (iii) find its area, (iv) write the coordinates of the point where its diagonals meet. *[Awareness · Coordinate Geometry]*

OR

b) i) Prove that the sum of the three angles of a triangle is 180°.
ii) The angles of a triangle are $(x + 10)°$, $(2x - 20)°$ and $(3x + 10)°$. Find the angles. *[Awareness · Lines and Angles]*

— ALL THE BEST —

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